

Developing AI Guidelines for Infrastructure-as-a-Service Platforms

RSAFRICA 2026

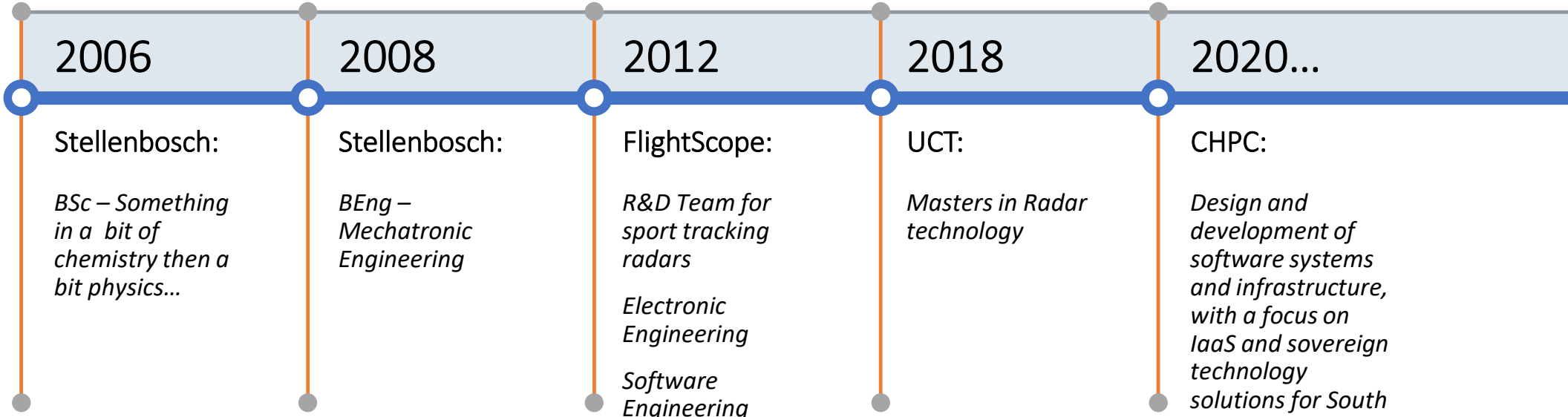
BINJAMIN BARSCH (BIN-YA-MEEN)

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

A bit about me…

Binjamin (Bin-ya-meen) Barsch

CHPC Lead Software Engineer



Spare time: Stop my kids from kitchen "experiments“!

CENTRE FOR HIGH PERFORMANCE COMPUTING

NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM

CHPC

NATIONAL SUPERCOMPUTING FACILITY

- 1PF system
- Formerly Top500
- Largest HPC System in Africa (YMMV)





... FOR RESEARCH & INDUSTRY

- South African Research Institutions & Government
- Southern Africa Community partners
- Industry representatives



science & innovation
Department: Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

Why AI Guidelines Matter…

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

science & innovation
Department:
Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

…too prevent this…

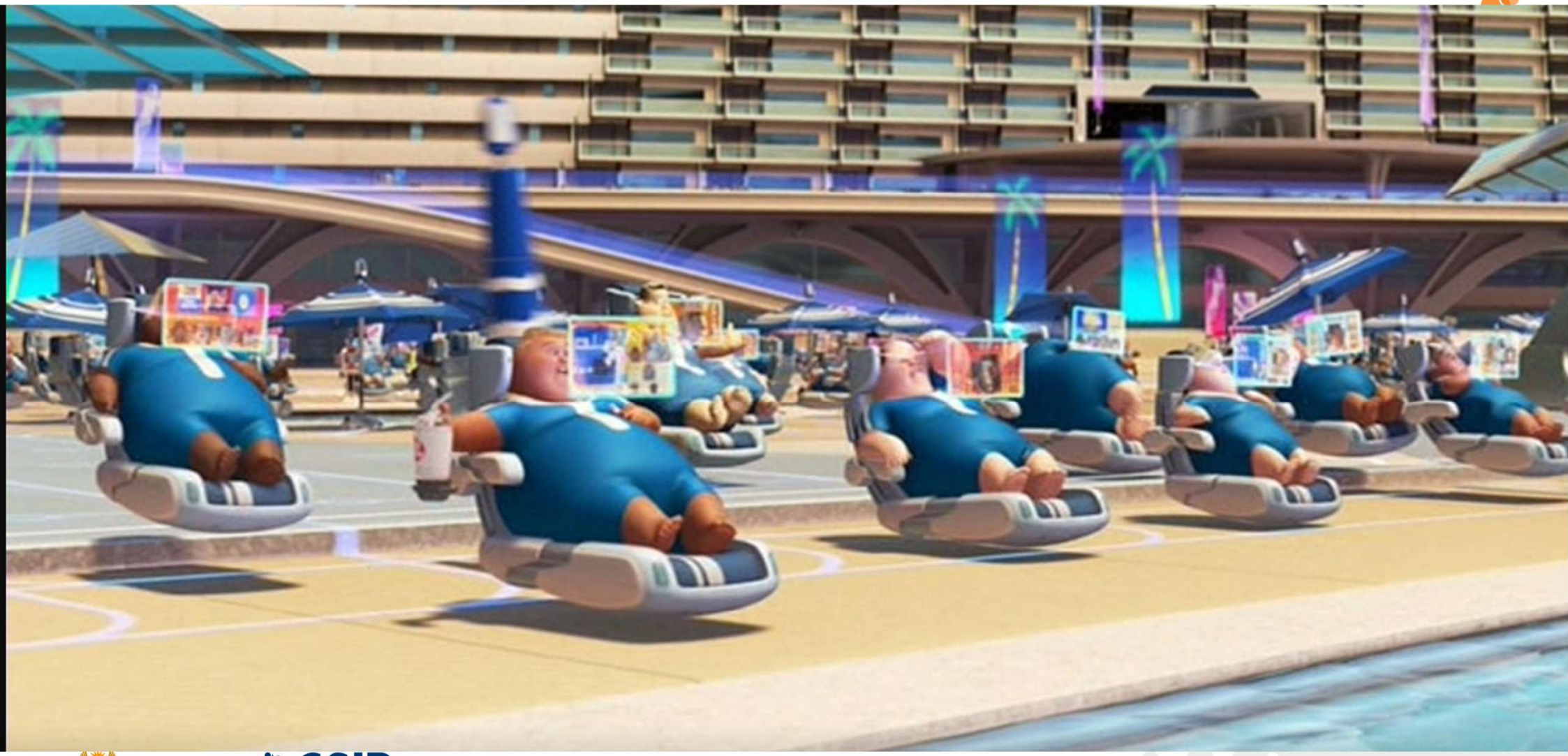
A national initiative of the Department of Science and Innovation and implemented by the CSIR.

science & innovation
Department:
Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation


CHRISTIAN BALE
SAM WORTHINGTON
TERMINATOR
SALVATION


FROM THE HUMANS WHO BROUGHT YOU "FINDING NEMO"
DISNEY · PIXAR
WALL·E
JUNE 27



Why AI Guidelines Matter

- AI assistants and coding copilots are everywhere
- AI touches infrastructure, not just code
- This introduces new operational and security risks
- Governance must become practical, not theoretical

“Insurance company chatbot… ☹”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

3 Levels of AI Autonomy

- Level 1 — Human-in-the-Loop (HITL)
- Level 2 — Human-on-the-Loop (HOTL)
- Level 3 — Human-out-of-the-Loop (HOOTL)

"AI is stochastic, not deterministic; not every answer is the same."

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Level 1 — Human-in-the-Loop

Human makes final decisions and executes actions.

- AI-assisted coding
- Documentation generation
- Debugging assistance
- Chatbots, ChatGPT, Copolit etc…
- These slides ☺

"Lowest risk. Human remains in control."

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Level 2 — Human-On-The-Loop

AI performs actions autonomously while humans monitor and intervene when needed.

- AI CI/CD assistants
- Automated cloud remediation
- AI infrastructure monitoring
- Cursor Agent, Claude Code etc…

“Moderate risk. Requires strong oversight.”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Level 3 — Human-Out-Of-the-Loop

AI operates independently with little or no human involvement.

- Fully autonomous agents
- Self-healing infrastructure
- AI-to-AI orchestration
- OpenHands, Open WebUI, …

“Highest risk. Dangerous in critical infrastructure.”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

3 Levels of AI Autonomy

- Level 1 — Human-in-the-Loop (HITL)
- Level 2 — Human-on-the-Loop (HOTL)
- Level 3 — Human-out-of-the-Loop (HOOTL)

"AI should improve a proven process, not replace an undefined one."

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

The Shift

From:

- AI as a chatbot

To:

- AI writing code
- AI running scripts
- AI accessing APIs
- AI controlling infrastructure

"YouTube Add – Don't use AI like you google…but I like to code !"

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

What is IaaS?

Infrastructure-as-a-Service includes (IaaS):

- Cloud virtual machines
- Storage systems
- Networks
- HPC clusters
- CI/CD systems
- Research platforms

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Two Sides of AI Governance

Shared Responsibility:

1. AI Users
2. AI Providers

Both need guardrails.

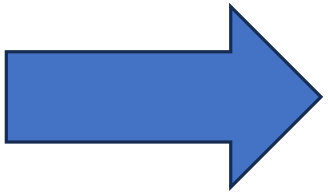
"Policies, governance, guidelines...sounds pretty boring!"

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

AI Users

Examples:

- Developers
- Sysadmins
- Researchers
- DevOps engineers



They use AI to:

- Write code
- Debug systems
- Automate tasks
- Query infrastructure

Risks from Users

Common risks:

- Credential leakage
- Blind trust in AI output (Biases)
- Unsafe scripts
- Wrong commands
- Over-permissioned agents
- …ability to think…

"…Can you replace AI"

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

AI Providers

Examples:

- Cloud providers
- HPC centres
- Platform teams
- Infrastructure operators

“They provide the environments AI can access.”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Risks from Provider Side

Examples:

- Excessive AI permissions
- Missing audit trails
- No sandboxing
- Weak policy controls
- Poor monitoring

“They provide the environments AI can access.”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Real Example

Prompt: “Fix my cluster issue”:

AI may:

- Kill jobs
- Delete files
- Modify configs
- Restart services
- Expose OS, Network vulnerabilities etc…

Fast ≠ Safe

South African AI Policy Incident

In 2026, South Africa withdrew its draft national AI policy after fictitious AI-generated citations were discovered.



NEWS SOUTH AFRICA

Minister Solly Malatsi's withdrawal of draft national AI policy for its fabricated sources 'embarrassing and damning'

Theolin Tembo | Published 2 weeks ago

4min

South African AI Policy Incident

Key lessons:

- Which level of AI was used?
- AI output is not verified truth
- Human review is mandatory
- Governance applies even to policymakers
- AI can accelerate mistakes at scale

"If AI can hallucinate inside AI policy itself, what happens when AI touches infrastructure?"

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

From Theory to Practice

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

What Are We Actually Doing?

At the CHPC, AI governance is approached in layers:

1. Organization-wide AI policy
2. Infrastructure-specific AI policy
3. Technical enforcement controls

"Policy alone is not enough. Controls must enforce policy"

CSIR AI Policy Foundation

Broad AI governance principles include:

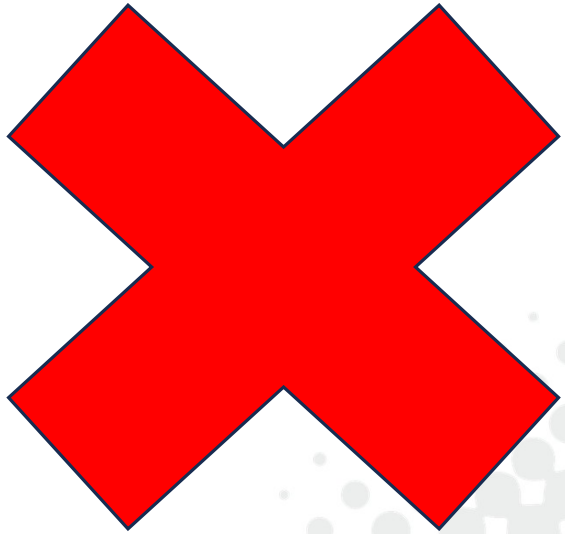
- Responsible and ethical AI use
- Human accountability remains mandatory
- Protection of sensitive data
- Legal compliance
- Transparency in AI-assisted work

“This provides the high-level governance framework.”

Why Generic AI Policy Is Not Enough

Infrastructure teams need more specific rules:

- Can AI access SSH credentials?
- Can AI trigger deployments?
- Can AI access production databases?
- Can AI execute privileged commands?



Infrastructure needs explicit guardrails.

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Data Sovereignty Matters

AI governance is not only about access control.

It is also about:

- Where data lives
- Where computation happens
- Which legal jurisdiction applies
- Who controls the infrastructure

“Key question…Where is your AI processing your data?”

Convenient ≠ Safe

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Data Sovereignty Matters

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

At CHPC, governance is not just theoretical.

We apply it through infrastructure:

- HPC cluster (Lengau)
- Local cloud (Sebowa)
- South African data centres (Cape Town and Pretoria)
- As a researcher, this service is for FREE!

This allows users to run workloads without exporting sensitive data offshore.

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

science & innovation
Department:
Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation

Local Infrastructure

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

With local HPC or cloud infrastructure, we can support:

- Data remaining in South Africa
- Better control over storage and access
- Compliance with local governance requirements
- Reduced exposure to foreign legal frameworks…geopolitics

Instead of sending data to a random foreign platform, we build infrastructure where controls are enforceable and you are in control!

CHPC AI Policy

Infrastructure-focused AI policy includes direct operational rules:

- No AI agent/tool may store or access private SSH keys
- No AI agent/tool may receive root or sudo credentials
- No autonomous production changes without approval
- No AI agent/tool may access sensitive user data by default
- All AI actions must be logged and auditable

These are easier to enforce for staff but more challenging for users

CHPC AI Policy

These controls can be directly enforced for infrastructure staff through technical and administrative controls.

For end users, complete enforcement is not possible!

Therefore, the infrastructure rules applies the following to everyone:

- Least-privilege access
- Network Isolation
- Monitoring and auditing
- Promoting secure AI usage through awareness

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Practical Application: Try Local Infrastructure!

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

A practical alternative is to run VMs, databases, or AI workloads **on local South African infrastructure.**

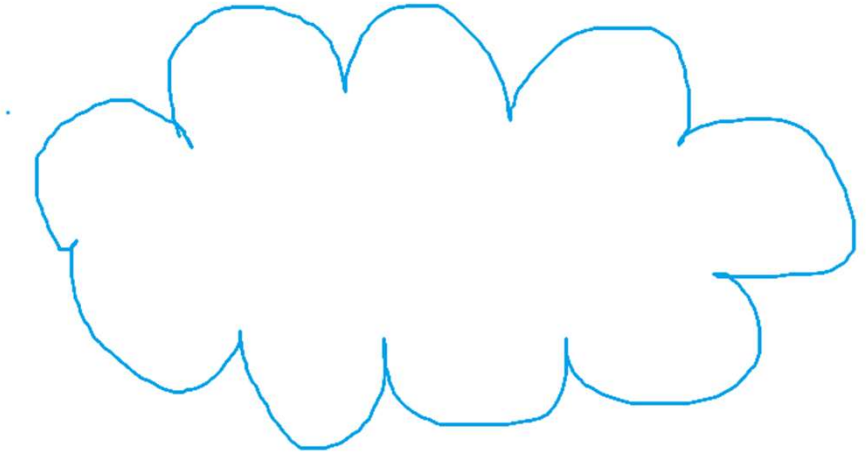
Through CHPC cloud infrastructure you have multiple options:

1. HPC Cluster Lengau – CPU and GPU Compute
2. Sebowa Cloud – Openstack Self Service Cloud Portal
3. VuthaStack – Spin up VM's and pre-built stacks with in minutes

A national initiative of the Department of Science and Innovation and implemented by the CSIR.
science & innovation
Department: Science and Innovation
REPUBLIC OF SOUTH AFRICA
CSIR
Touching lives through innovation

Sebowa Cloud

- Self-service Openstack cloud portal to launch and manage VMs.
- Networking and floating IPs.
- Security groups and SSH keys.
- Less restrictions to run AI agents
- South African hosted infrastructure.



“Link at the end of presentation”

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

VuthaStack: VMs in minutes

NICIS
NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM
CHPC

Quickly evaluate services before fully comprehensive Sebowa Openstack environment

Examples:

- VM Only: Ubuntu, Rocky Linux, CentOS VM
- GPU-enabled (NVIDIA A100) VM, and development sandbox
- Pre-built stacks: Chatbot, JupyterLab, Kubernetes
- Open WebUI (Qwen3.6, gemma4, qwen3)

This gives users hands-on access to South African infrastructure in a controlled environment.

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

science & innovation
Department:
Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation

Closing

Responsible AI infrastructure requires three pillars:

- Governance: what AI is allowed to do
- Security: what AI is allowed to access
- Sovereignty: where AI is allowed to run

Through supporting local infrastructure, AI governance becomes practical - not just policy.

In infrastructure, “trust” must be earned — never assumed.

A national initiative of the Department of Science and Innovation and implemented by the CSIR.



NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM

CHPC

Thank You!

Developing AI Guidelines for Infrastructure-as-a-Service Platforms – RSAfrica 2026

bbarsch@csir.co.za

science & innovation
Department:
Science and Innovation
REPUBLIC OF SOUTH AFRICA

CSIR
Touching lives through innovation